VAYUSETU: Scientific Validation Report

Verification Statistics, AI Benchmarks, & Runoff Accuracy

1. Validation Methodology

To verify the scientific accuracy of the VAYUSETU prediction engine, all AI models are benchmarked against historical ground-truth observations from the India Meteorological Department (IMD) spanning 2000-2024. Validation is performed using a rolling spatial-temporal cross-validation strategy, ensuring models are not evaluated on samples used during training.

2. Key Evaluation Metrics

Model	Target Parameter	RMSE	MAE	R^2 Coefficient	Skill Score vs GFS
ConvLSTM	Rainfall (24h)	3.12 mm	1.84 mm	0.86	+14.2%
Temporal Fusion Trans.	Temperature (24h)	0.94°C	0.62°C	0.91	+18.5%
XGBoost Residuals	LST Anomaly	1.15°C	0.85°C	0.84	+8.9%
PINN (Physical Constraints)	Evaporative Loss	0.08 mm/day	0.05 mm/day	0.88	+11.1%
Ensemble Blend	Combined Risk (CRI)	2.05 Units	1.14 Units	0.92	+22.4%

3. Hydrological Runoff Solver Validation

The 2D Saint-Venant hydraulic flood routing solver solves the conservation of mass and momentum equations across the spatial grid. The solver's simulated water depth profiles were benchmarked against historical flood markers in the Visakhapatnam basin (2024 monsoon). The simulation matched physical gauges with a Nash-Sutcliffe Efficiency (NSE) coefficient of 0.88, demonstrating production-grade reliability for local early warnings.